

DEVELOPER RUNBOOK

ReliSoft HR Portal

Step-by-Step Local Setup, Database Migrations, and Application Startup

Windows development environment

Prepared 19 August 2026 | Repository runbook

Outcome - At the end of this guide, LocalDB is migrated, the .NET API runs on port 5049, the Vite client runs on port 5173, Swagger is available, and a demo login is verified.

1. Overview

This runbook is the repeatable local-development procedure for the ReliSoft HR Portal. Run commands from Windows PowerShell or Command Prompt. Examples use the current workspace path; replace it if your clone is stored elsewhere.

System at a glance

Component	Local address or resource
Frontend	http://localhost:5173
Backend API	http://localhost:5049
Swagger UI	http://localhost:5049/swagger/index.html
Swagger JSON	http://localhost:5049/swagger/v1/swagger.json
Database	SQL Server LocalDB: RelisoftHRDb
Connection	Server=(localdb)\MSSQLLocalDB; Database=RelisoftHRDb; Integrated Security=true

Prerequisites

Windows 10 or Windows 11 with SQL Server LocalDB installed.

.NET SDK 10.x. The project targets net10.0.

Entity Framework Core CLI 10.0.9 or a compatible 10.x version.

Node.js and npm. This workspace was verified with Node 24.16.0 and npm 11.13.0.

Git is recommended for cloning, pulling changes, and reviewing migrations.

Verify the toolchain:

```
dotnet --version
dotnet ef --version
node --version
npm.cmd --version
sqllocaldb versions
```

PowerShell note - If Windows blocks npm.ps1 with an execution-policy error, use npm.cmd exactly as shown in this guide. No execution-policy change is required.

2. First-Time Installation

Step 1 - Open the repository root

```
cd E:\LMS\Relisoft-HR-Portal\relisoft-hr
```

The repository root contains package.json, server, client, README.md, and TECHNICAL.md.

Step 2 - Install root JavaScript dependencies

```
npm.cmd install
```

This installs the root development runner, including concurrently, which is used by npm run dev.

Step 3 - Restore backend packages

```
dotnet restore server\RelisoftHR.csproj
```

Step 4 - Install frontend dependencies

```
npm.cmd install --prefix client
```

Step 5 - Confirm configuration

File	Purpose
server/appsettings.json	LocalDB connection string, JWT settings, CORS, email, and leave policy.
server/appsettings.Development.json	Development logging configuration.
server/Properties/launchSettings.json	Backend development URL: http://localhost:5049.
client/vite.config.js	Frontend dev server and /api proxy to the backend.
package.json	Root commands for both services, database update, and builds.

Security - The checked-in JWT key and demo password are development defaults. Replace secrets through environment variables or secure production configuration before deployment.

3. Database Setup and Migrations

Step 1 - Check and start LocalDB

```
sqllocaldb info MSSQLLocalDB  
sqllocaldb start MSSQLLocalDB
```

A healthy instance reports State: Running and shows an instance pipe name. Run the application and LocalDB under the same Windows user account.

Step 2 - Apply all Entity Framework migrations

```
dotnet ef database update  
--project server\RelisoftHR.csproj  
--startup-project server\RelisoftHR.csproj
```

Expected result - The command ends with Done. The RelisoftHRDb database and the __EFMigrationsHistory table are created or updated.

Step 3 - Inspect migration status

```
dotnet ef migrations list  
--project server\RelisoftHR.csproj  
--startup-project server\RelisoftHR.csproj
```

The backend also calls Database.Migrate() during startup, so unapplied migrations are normally applied automatically. Running database update explicitly is recommended on first setup because it surfaces migration errors before the API starts.

Creating a new migration

```
dotnet ef migrations add DescriptiveMigrationName  
--project server\RelisoftHR.csproj  
--startup-project server\RelisoftHR.csproj  
  
dotnet ef database update  
--project server\RelisoftHR.csproj  
--startup-project server\RelisoftHR.csproj
```

Database safety - Do not run database drop, delete MDF/LDF files, or recreate a database that may contain user data. Back up first and obtain explicit approval for destructive recovery.

4. Start the Application

Option A - Run backend and frontend separately (recommended)

Terminal 1 - backend:

```
cd E:\LMS\Relisoft-HR-Portal\relisoft-hr
dotnet run --project server\RelisoftHR.csproj
```

Wait for: Now listening on: http://localhost:5049

Terminal 2 - frontend:

```
cd E:\LMS\Relisoft-HR-Portal\relisoft-hr
npm.cmd run dev --prefix client
```

Wait for Vite to report its Local URL, normally http://localhost:5173.

Option B - Run both from one terminal

```
cd E:\LMS\Relisoft-HR-Portal\relisoft-hr
npm.cmd run dev
```

The root dev script uses concurrently and labels the backend and frontend output separately.

Stop or restart

Press Ctrl+C in each terminal to stop a separately launched service.

For the combined runner, press Ctrl+C once and confirm termination if prompted.

After changing C# code, rebuild or allow dotnet run to rebuild. After changing Vite code, hot reload is normally automatic.

Port ownership - Before force-stopping a process, verify its executable and port. Avoid terminating unrelated Node or .NET processes.

5. Verify the Running System

Browser checks

Open <http://localhost:5173> and confirm that the ReliSoft login screen loads.

Open <http://localhost:5049/swagger/index.html> and confirm that Swagger UI lists the API endpoints.

In Swagger, expand POST /api/auth/login, choose Try it out, submit the demo credentials, and confirm HTTP 200.

Demo login

Username	Password	Typical role
preeti	password	HR L2
rakesh	password	Organization Head
aradhana	password	Employee

The database must contain the corresponding demo seed data. The preeti account is the recommended first verification account.

PowerShell API health check

```
$body = @{{
  username = 'preeti'
  password = 'password'
} | ConvertTo-Json

Invoke-RestMethod
  -Uri 'http://localhost:5049/api/auth/login'
  -Method Post
  -ContentType 'application/json'
  -Body $body
```

Success signal - A successful response contains the username, employee information, role, permitted views, and a JWT token.

Listening-port check

```
Get-NetTCPConnection -State Listen |
  Where-Object { $_.LocalPort -in 5049, 5173 } |
  Select-Object LocalAddress, LocalPort, OwningProcess
```

6. Build, Test, and Daily Workflow

Build both applications

```
dotnet build server\RelisoftHR.csproj --no-restore  
npm.cmd run build --prefix client
```

Run automated checks

```
dotnet test server\RelisoftHR.Tests\RelisoftHR.Tests.csproj  
npm.cmd run test --prefix client  
npm.cmd run lint --prefix client
```

Recommended workflow after pulling changes

Open the repository root and review git status so local work is not overwritten.

Restore backend and frontend dependencies if package files changed.

Run dotnet ef database update before starting the API when migrations changed.

Build the backend and frontend.

Start both services and verify Swagger plus the login endpoint.

Quick acceptance checklist

LocalDB MSSQLLocalDB is running.

All EF migrations are applied without errors.

Backend responds on port 5049.

Swagger UI and swagger.json both return HTTP 200.

Frontend responds on port 5173.

preeti / password returns HTTP 200 from POST /api/auth/login.

Backend and frontend builds complete successfully.

7. Troubleshooting

LocalDB cannot create or discover MSSQLLocalDB

Symptoms include SQL Network Interfaces error 50, Cannot create an automatic instance, or an error reading the LocalDB registry configuration.

```
sqllocaldb info
sqllocaldb info MSSQLLocalDB
sqllocaldb start MSSQLLocalDB
```

Run the API and LocalDB as the same Windows user.

Confirm the connection string targets (localdb)\MSSQLLocalDB.

Repair SQL Server LocalDB if instance metadata remains inaccessible. Do not delete the database files as a first response.

Database update fails on ManagerCode foreign key

Use the current repository migrations. They normalize manager references that point to demo employees not yet inserted, recreate the self-reference safely, and assign managers after the referenced employees exist.

```
git status
dotnet build server\RelisoftHR.csproj
dotnet ef database update
--project server\RelisoftHR.csproj
--startup-project server\RelisoftHR.csproj
```

Invalid column name during startup

The application model and database schema are out of sync. Pull the latest migration corrections, rebuild, and apply database update. Never hide this error by removing model properties.

Swagger is missing or returns an error

Use /swagger/index.html, not the older /openapi/v1.json route.

Confirm ASPNETCORE_ENVIRONMENT is Development; Swagger UI is intentionally development-only.

Rebuild and restart the backend after changing Program.cs or package references.

Troubleshooting (continued)

npm.ps1 is blocked

Use npm.cmd in PowerShell:

```
npm.cmd install
npm.cmd run dev --prefix client
```


Port 5049 or 5173 is already in use

```
Get-NetTCPConnection -State Listen |
  Where-Object { $_.LocalPort -in 5049, 5173 } |
  Select-Object LocalPort, OwningProcess

Get-Process -Id <OwningProcess>
```

Stop the existing ReliSoft process cleanly with Ctrl+C. Use Stop-Process only after confirming that the PID belongs to this project.

Frontend shows Failed to fetch

Confirm the backend is running on <http://localhost:5049>.

Confirm client/vite.config.js proxies /api requests to port 5049.

Check the backend terminal for SQL, migration, or seeding errors.

Windows Event Log access error

Development configuration disables the Windows Event Log provider. Confirm `server/appsettings.Development.json` is present and `ASPNETCORE_ENVIRONMENT` is Development.

8. Command Reference

Task	Command
Install root	<code>npm.cmd install</code>
Restore backend	<code>dotnet restore server\RelisoftHR.csproj</code>
Install client	<code>npm.cmd install --prefix client</code>
Apply migrations	<code>npm.cmd run database-update</code>
Run both	<code>npm.cmd run dev</code>
Run backend	<code>dotnet run --project server\RelisoftHR.csproj</code>
Run frontend	<code>npm.cmd run dev --prefix client</code>
Build all	<code>npm.cmd run build</code>
Swagger	http://localhost:5049/swagger/index.html

Ready - When all acceptance checks pass, keep both service terminals open and develop against <http://localhost:5173>. Use Swagger to inspect and exercise backend endpoints.